

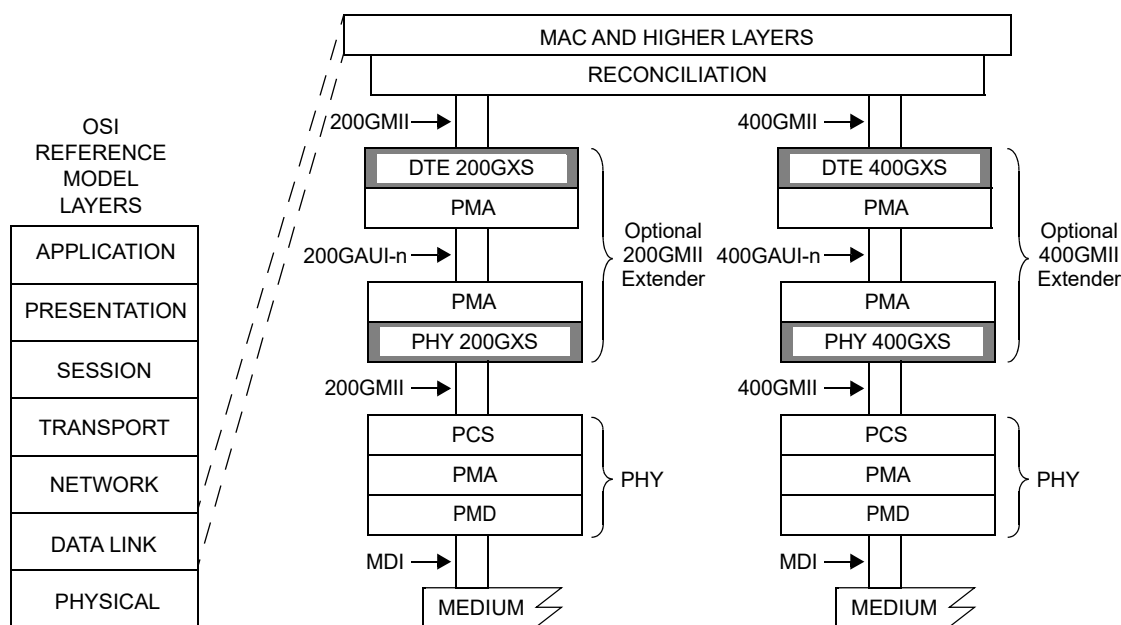
118. 200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS)

118.1 Overview

This clause defines the functional characteristics for the optional 200GMII Extender and 200GMII Extender Sublayer (200GXS), and also for the optional 400GMII Extender and 400GMII Extender Sublayer (400GXS). Figure 118–1 shows the relationship of the 200GMII/400GMII Extender and 200GXS/400GXS sublayer with other sublayers to the ISO Open System Interconnection (OSI) reference model.

The 200GMII/400GMII Extender allows the extension of the 200GMII/400GMII to the PCS via a physical instantiation. The 200GMII/400GMII Extender is composed of a 200GXS/400GXS at the RS end, a 200GXS/400GXS at the PHY end with a physical instantiation of 200GAUI-n/400GAUI-n between two adjacent PMA sublayers.

A 200GMII/400GMII Extender with the optional Energy-Efficient Ethernet (EEE) capability (see Clause 78) encodes and decodes Low Power Idle (LPI) signals. The assertion of LPI at the 200GMII/400GMII is encoded in the transmitted symbols. Detection of LPI encoding in the received symbols is indicated as LPI at the 200GMII/400GMII.



200GAUI-n = 200 Gb/s n-LANE ATTACHMENT UNIT
INTERFACE

200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE

200GXS = 200GMII EXTENDER SUBLAYER

400GAUI-n = 400 Gb/s n-LANE ATTACHMENT UNIT
INTERFACE

400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE

400GXS = 400GMII EXTENDER SUBLAYER

DTE = DATA TERMINAL EQUIPMENT

MAC = MEDIA ACCESS CONTROL

MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE

PMA = PHYSICAL MEDIUM ATTACHMENT

PMD = PHYSICAL MEDIUM DEPENDENT

Figure 118–1—200GXS and 400GXS relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

118.1.1 Summary of major concepts

The following is a list of the major concepts of the 200GMII/400GMII Extender:

- a) Simple signal mapping to the 200GMII/400GMII
- b) The optional 200GMII/400GMII Extender can be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 200GMII/400GMII
- c) Independent transmit and receive data paths
- d) The 200GXS/400GXS leverages all functions in the Clause 119 PCS and supports physical instantiations of the 200GAUI-n/400GAUI-n
- e) Optionally extends LPI signaling to the PHY for EEE

118.1.2 200GXS/400GXS Sublayer

The 200GXS, if implemented, shall be identical in function to the 200GBASE-R PCS in Clause 119 with the addition of the functions defined in 118.2. A single device may be configured as either a 200GXS or the 200GBASE-R PCS and may be managed through different optional management registers.

The 400GXS, if implemented, shall be identical in function to the 400GBASE-R PCS in Clause 119 with the addition of the functions defined in 118.2. A single device may be configured as either a 400GXS or the 400GBASE-R PCS and may be managed through different optional management registers.

118.1.3 200GAUI-n/400GAUI-n

A 200GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers (200GAUI-n), as shown in Figure 118–1:

- 200GAUI-8 chip-to-chip (Annex 120B)
- 200GAUI-8 chip-to-module (Annex 120C)
- 200GAUI-4 chip-to-chip (Annex 120D)
- 200GAUI-4 chip-to-module (Annex 120E)

A 400GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers (400GAUI-n), as shown in Figure 118–1:

- 400GAUI-16 chip-to-chip (Annex 120B)
- 400GAUI-16 chip-to-module (Annex 120C)
- 400GAUI-8 chip-to-chip (Annex 120D)
- 400GAUI-8 chip-to-module (Annex 120E)

118.2 FEC Degrade

The propagation of FEC degrade signaling across PCS and XS sublayers is described in 116.6 and is based on the optional FEC degrade signaling described in Clause 119 with the changes described for the DTE XS in 118.2.1 and for the PHY XS in 118.2.2.

118.2.1 DTE XS FEC Degrade signaling

The variable `tx_am_sf` is set as follows:

`tx_am_sf<2:0> = {FEC_degraded_SER + rx_local_degraded,0,0}`

118.2.2 PHY XS FEC Degradе signaling

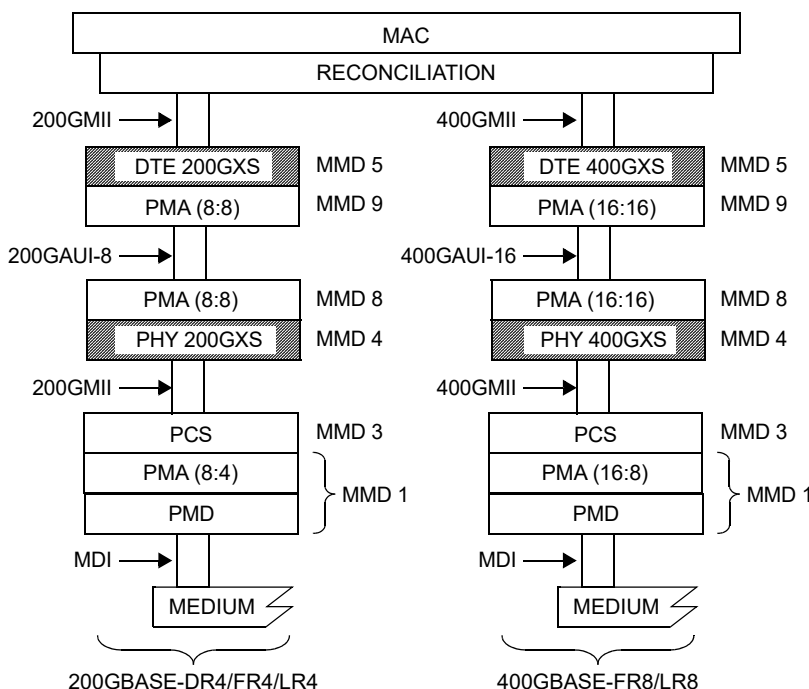
The variable `tx_am_sf` is set as follows:

$$\text{tx_am_sf} < 2:0 > = \{\text{PCS:rx_rm_degraded}, \text{PCS:FEC_degraded_SER} + \text{PCS:rx_local_degraded}, 0\}$$

Where `PCS:rx_rm_degraded`, `PCS:FEC_degraded_SER`, and `PCS:rx_local_degraded` are the `rx_rm_degraded`, `FEC_degraded_SER`, and `rx_local_degraded` variables from the adjacent PCS.

118.3 200GXS and 400GXS partitioning example

A partitioning example and MMD numbering using the 200GXS and 400GXS is shown in Figure 118–2.



200GAUI = 200 Gb/s ATTACHMENT UNIT INTERFACE
200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
200GXS = 200 Gb/s EXTENDER SUBLAYER
400GAUI = 400 Gb/s ATTACHMENT UNIT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
400GXS = 400 Gb/s EXTENDER SUBLAYER
DTE = DATA TERMINAL EQUIPMENT

MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
MMD = MDIO MANAGEABLE DEVICE
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 118–2—Example 200GBASE-DR4/FR4/LR4 and 400GBASE-FR8/LR8 PMA layering with 200GXS and 400GXS

118.4 200GXS and 400GXS MDIO function mapping

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for and about the 200GXS or 400GXS. If MDIO is implemented, it shall map MDIO PHY XS and DTE XS control bits to Clause 119 control variables as shown in Table 118–1 and Table 118–3, respectively. Similarly, if MDIO is implemented, it shall map MDIO PHY XS and DTE XS status bits to Clause 119 status variables as shown in Table 118–2 and Table 118–4, respectively.

Table 118–1—MDIO PHY XS to Clause 119 control variable mapping

MDIO control variable	PHY XS register name	Register/bit number	Clause 119 control variable
Reset	PHY XS control 1 register	4.0.15	reset
Loopback	PHY XS control 1 register	4.0.14	Loopback
Transmit test-pattern enable	BASE-R PHY XS test-pattern control register	4.42.3	tx_test_mode
PHY XS FEC bypass indication enable	PHY XS FEC control register	4.800.1	FEC_bypass_indication_enable
PHY XS FEC degraded SER enable	PHY XS FEC control register	4.800.2	FEC_degraded_SER_enable
PHY XS FEC degraded SER activate threshold	PHY XS FEC degraded SER activate threshold register	4.806, 4.807	FEC_degraded_SER_activate_threshold
PHY XS FEC degraded SER deactivate threshold	PHY XS FEC degraded SER deactivate threshold register	4.808, 4.809	FEC_degraded_SER_deactivate_threshold
PHY XS FEC degraded SER interval	PHY XS FEC degraded SER interval	4.810, 4.811	FEC_degraded_SER_interval

Table 118–2—MDIO PHY XS to Clause 119 status variable mapping

MDIO status variable	PHY XS register name	Register/bit number	Clause 119 status variable
BASE-R PHY XS receive link status	BASE-R PHY XS status 1	4.32.12	PCS_status
Lane <i>x</i> aligned	Multi-lane BASE-R PHY XS alignment status 3 and 4	4.52.7:0 4.53.7:0	am_lock< <i>x</i> >
PHY XS lane alignment status	Multi-lane BASE-R PHY XS alignment status 1	4.50.12	align_status
Lane <i>x</i> mapping	PHY XS lane mapping, lane 0 through lane 15	4.400 through 4.415	pcs_lane_mapping< <i>x</i> >
PHY XS FEC bypass indication ability	PHY XS FEC status	4.801.1	FEC_bypass_indication_ability
FEC corrected codewords	PHY XS FEC corrected codewords counter	4.802, 4.803	FEC_corrected_cw_counter
FEC uncorrected codewords	PHY XS FEC uncorrected codewords counter	4.804, 4.805	FEC_uncorrected_cw_counter
PHY XS FEC symbol errors, lane 0 to lane 15	PHY XS FEC symbol error counter, lane 0 to lane 15	4.600 to 4.631	FEC_symbol_error_counter_ <i>i</i>
Tx LPI indication	PHY XS status 1	4.1.9	Tx LPI indication
Tx LPI received	PHY XS status 1	4.1.11	Tx LPI received
Rx LPI indication	PHY XS status 1	4.1.8	Rx LPI indication
Rx LPI received	PHY XS status 1	4.1.10	Rx LPI received
EEE wake error counter	EEE wake error counter	4.22	Wake_error_counter

Table 118–2—MDIO PHY XS to Clause 119 status variable mapping (*continued*)

MDIO status variable	PHY XS register name	Register/bit number	Clause 119 status variable
PHY XS FEC degraded SER ability	PHY XS FEC status register	4.801.3	FEC_degraded_SER_ability
PHY XS FEC degraded SER	PHY XS FEC status register	4.801.4	FEC_degraded_SER
Remote degraded SER received	PHY XS FEC status register	4.801.5	rx_rm_degraded

Table 118–3—MDIO DTE XS to Clause 119 control variable mapping

MDIO control variable	DTE XS register name	Register/bit number	Clause 119 control variable
Reset	DTE XS control 1 register	5.0.15	reset
Loopback	DTE XS control 1 register	5.0.14	Loopback
Transmit test-pattern enable	BASE-R DTE XS test-pattern control register	5.42.3	tx_test_mode
DTE XS FEC bypass indication enable	DTE XS FEC control register	5.800.1	FEC_bypass_indication_enable
DTE XS FEC degraded SER enable	DTE XS FEC control register	5.800.2	FEC_degraded_SER_enable
DTE XS FEC degraded SER activate threshold	DTE XS FEC degraded SER activate threshold register	5.806, 5.807	FEC_degraded_SER_activate_threshold
DTE XS FEC degraded SER deactivate threshold	DTE XS FEC degraded SER deactivate threshold register	5.808, 5.809	FEC_degraded_SER_deactivate_threshold
DTE XS FEC degraded SER interval	DTE XS FEC degraded SER interval	5.810, 5.811	FEC_degraded_SER_interval

Table 118–4—MDIO DTE XS to Clause 119 status variable mapping

MDIO status variable	DTE XS register name	Register/bit number	Clause 119 status variable
BASE-R DTE XS receive link status	BASE-R DTE XS status 1	5.32.12	PCS_status
Lane <i>x</i> aligned	Multi-lane BASE-R DTE XS alignment status 3 and 4	5.52.7:0 5.53.7:0	am_lock< <i>x</i> >
DTE XS lane alignment status	Multi-lane BASE-R DTE XS alignment status 1	5.50.12	align_status
Lane <i>x</i> mapping	DTE XS lane mapping, lane 0 through lane 15	5.400 through 5.415	pcs_lane_mapping< <i>x</i> >
DTE XS FEC bypass indication ability	DTE XS FEC status	5.801.1	FEC_bypass_indication_ability
FEC corrected codewords	DTE XS FEC corrected codewords counter	5.802, 5.803	FEC_corrected_cw_counter

Table 118–4—MDIO DTE XS to Clause 119 status variable mapping (*continued*)

MDIO status variable	DTE XS register name	Register/bit number	Clause 119 status variable
FEC uncorrected codewords	DTE XS FEC uncorrected codewords counter	5.804, 5.805	FEC_uncorrected_cw_counter
DTE XS FEC symbol errors, lane 0 to lane 15	DTE XS FEC symbol error counter, lane 0 to lane 15	5.600 to 5.631	FEC_symbol_error_counter_ <i>i</i>
Tx LPI indication	DTE XS status 1	5.1.9	Tx LPI indication
Tx LPI received	DTE XS status 1	5.1.11	Tx LPI received
Rx LPI indication	DTE XS status 1	5.1.8	Rx LPI indication
Rx LPI received	DTE XS status 1	5.1.10	Rx LPI received
EEE wake error counter	EEE wake error counter	5.22	Wake_error_counter
DTE XS FEC degraded SER ability	DTE XS FEC status register	5.801.3	FEC_degraded_SER_ability
DTE XS FEC degraded SER	DTE XS FEC status register	5.801.4	FEC_degraded_SER
Remote degraded received	DTE XS FEC status register	5.801.5	rx_rm_degraded
Local degraded received	DTE XS FEC status register	5.801.6	rx_local_degraded

118.5 Protocol implementation conformance statement (PICS) proforma for Clause 118, 200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS)²⁰⁵

118.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 118, 200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

118.5.2 Identification

118.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

118.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 118, 200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	
Date of Statement	

²⁰⁵ *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

118.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
MII	200GMII or 400GMII logical interface	117, 118.1	Logical interface is supported	O	Yes [] No []
*PHYXS	PHY 200GXS or PHY 400GXS	118.1		O/2	Yes [] No []
*DTEXS	DTE 200GXS or DTE 400GXS	118.1		O/2	Yes [] No []
*200GXS	Using the XS for 200GBASE-R	118.1.2		O.1	Yes [] No []
*400GXS	Using the XS for 400GBASE-R	118.1.2		O.1	Yes [] No []
*MD	MDIO	45, 118.4	Registers and interface supported	O	Yes [] No []
*BI	Bypass indication	119.2.5.3	Capability is supported	O	Yes [] No []
DC	Delay constraints	119.5	Conforms to delay constraints specified in 119.5	M	Yes []
EEE	EEE capability	119.2.3.3	Capability is supported	O	Yes [] No []
JTM	Supports test-pattern mode	119.2.1, 119.2.4.9		M	Yes []
FDD	Support for optional FEC degraded detection	119.2.5.3	In the FEC decoder can optionally detect FEC SER degraded at a programmable threshold	O	Yes [] No []

118.5.4 PICS proforma tables for 200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS)

118.5.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	64B/66B to 256B/257B transcoder	119.2.4.2	tx_xcoded<256:0> constructed per 119.2.4.2	M	Yes []
TF2	Transmission bit ordering	119.2.4.8	First bit transmitted is bit 0	M	Yes []
TF3	Pad value	119.2.4.4	PRBS9	M	Yes []
TF4	Alignment marker insertion	119.2.4.4		M	Yes []
TF5	Pre-FEC distribution	119.2.4.5	Distribute the data to two FEC codewords	M	Yes []
TF6	Reed-Solomon encoder	119.2.4.6	RS(544,514)	M	Yes []
TF7	Symbol distribution	119.2.4.7	Distribution is based on 10b symbols	M	Yes []

118.5.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Skew tolerance	119.2.5.1	Maximum Skew of 180 ns between PCS lanes and a maximum Skew Variation of 4 ns	M	Yes []
RF2	Lane reorder and de-interleave	119.2.5.2	Order the PCS lanes according to the PCS lane number and de-interleave the FEC codewords	M	Yes []
RF3	Reed-Solomon decoder	119.2.5.3	Corrects any combination of up to $t=15$ symbol errors in a codeword	M	Yes []
RF4	Reed-Solomon decoder	119.2.5.3	Capable of indicating when a codeword was not corrected.	M	Yes []
RF5	Error monitoring while error indication is bypassed	119.2.5.3	When the number of symbol errors in a block of 8192 codewords exceeds 5560 assert hi_ser for 60 ms to 75 ms	BI:M	Yes [] N/A []
RF6	256B/257B to 64B/66B transcoder	119.2.5.7	rx_coded_j<65:0>, j=0 to 3 constructed per 119.2.5.7	M	Yes []

118.5.4.3 64B/66B coding rules

Item	Feature	Subclause	Value/Comment	Status	Support
C1	Encoder (and ENCODE function) implements the code as specified	119.2.3, 119.2.6.2.3		M	Yes []
C2	Decoder (and DECODE function) implements the code as specified	119.2.3, 119.2.6.2.3		M	Yes []
C3	Only valid block types are transmitted	119.2.3.2		M	Yes []
C4	Invalid block types are treated as an error	119.2.3.2		M	Yes []
C5	Only valid control characters are transmitted	119.2.3.3		M	Yes []
C6	Invalid control characters are treated as an error	119.2.3.3		M	Yes []
C7	Idles do not interrupt data	119.2.3.5		M	Yes []
C8	IDLE control code insertion and deletion	119.2.3.5	Insertion or Deletion in groups of 8 I/s	M	Yes []
C9	Sequence ordered set deletion	119.2.3.8	Only one whole ordered set of two consecutive sequence ordered sets may be deleted	M	Yes []

118.5.4.4 Scrambler and descrambler

Item	Feature	Subclause	Value/Comment	Status	Support
S1	Scrambler	119.2.4.3	Performs as shown in Figure 49–8	M	Yes []
S2	Descrambler	119.2.5.6	Performs as shown in Figure 49–10	M	Yes []

118.5.4.5 Alignment markers

Item	Feature	Subclause	Value/Comment	Status	Support
AM1	Alignment marker insertion	119.2.4.4	Alignment markers are inserted periodically as in 119.2.4.4	M	Yes []
AM2	Alignment marker form	119.2.4.4	Alignment markers are formed as described in 119.2.4.4	M	Yes []
AM3	Lane mapping	119.2.4.4	PCS lane number is captured	MD:M	Yes [] N/A []
AM4	Alignment marker removal	119.2.5.5	Alignment markers are removed prior to descrambling as described in 119.2.5.5	M	Yes []

118.5.5 Test-pattern modes

Item	Feature	Subclause	Value/Comment	Status	Support
JT1	Scrambled idle transmit test-pattern generator is implemented	119.2.4.9		M	Yes []

118.5.6 Bit order

Item	Feature	Subclause	Value/Comment	Status	Support
B1	Transmit bit order	119.2.4.8	Placement of bits into the PCS lanes as shown in Figure 119–10 or Figure 119–11	M	Yes []

118.5.7 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Equivalent access to XS management objects is provided	119.3		!MD:O	Yes [] No [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits for PHY 200GXS or PHY 400GXS	118.4	See Table 118–1 and Table 118–2	MD*PHY XS:M	Yes [] N/A []
M3	Mapping of MDIO control bits and MDIO status bits for DTE 200GXS or DTE 400GXS	118.4	See Table 118–3 and Table 118–4	MD*DTEX S:M	Yes [] N/A []

118.5.7.1 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	Alignment marker lock	119.2.6	Implements 8 alignment marker lock processes as depicted in Figure 119–12	200GXS: M	Yes [] N/A []
SM2	Alignment marker lock	119.2.6	Implements 16 alignment marker lock processes as depicted in Figure 119–12	400GXS: M	Yes [] N/A []
SM3	The SLIP function evaluates all possible block positions	119.2.6.2.3		M	Yes []
SM4	PCS synchronization state diagram	119.2.6	Meets the requirements of Figure 119–13	M	Yes []
SM5	Transmit process	119.2.6	Meets the requirements of Figure 119–14	M	Yes []
SM6	Receive process	119.2.6	Meets the requirements of Figure 119–15	M	Yes []

118.5.7.2 Loopback

Item	Feature	Subclause	Value/Comment	Status	Support
L1	Supports loopback	119.4		M	Yes []
L2	When in loopback, transmits what it receives from the 200GMII/400GMII	119.4		M	Yes []
L3	When in loopback, ignore all data presented by the PMA sublayer	119.4		M	Yes []

118.5.7.3 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
TIM1	PCS delay constraint	119.5	No more than 160 256 BT for sum of transmit and receive path delays for 200GBASE-R.	200GXS :M	Yes [] N/A []
TIM2	PCS delay constraint	119.5	No more than 320 000 BT for sum of transmit and receive path delays for 400GBASE-R.	400GXS :M	Yes [] N/A []